

Differential Pairs

Conductor Width (W)  
**1** mils

Conductor Spacing (S)  
**5** mils

Conductor Height (H1)  
**3.14961** mils

Conductor Height (H2)  
**0.98425** mils

Applied Voltage  
**1.2** Volts

Coupled Length  
**21** mils

Signal Risetime  
**0.090** ns

Target Zdiff  
**100** Ohms

+/- Tolerance = 15%  


Target Zdiff Plus  
**115.000** Ohms

Target Zdiff Minus  
**85.000** Ohms

Formula Restrictions:  
0.1 < W/H < 3.0  
0.1 < S/H < 3.0

Differential Protocol  
**DDR4 CLK/DQS**

Send to Via Calculator

Zdifferential  
**96.229** Ohms

Zo  
**49.461** Ohms

Zodd  
**48.114** Ohms

Zeven  
**50.845** Ohms

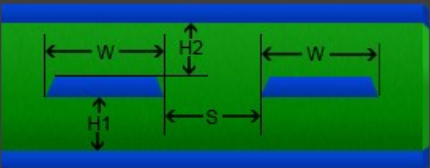
? Help

Kb (Term)  
**0.0138** dB

Kb' (Unterm)  
**0.0276** dB

NEXT Voltage  
**0.0014** V

NEXT Voltage  
**0.0028** V



Options

Base Copper Weight  
☐ 0.25oz  
☐ 0.5oz  
☒ 1oz  
☐ 1.5oz  
☐ 2oz  
☐ 2.5oz  
☐ 3oz  
☐ 4oz  
☐ 5oz

Plating Thickness  
☐ Bare PCB  
☐ 0.5oz  
☐ 1oz  
☐ 1.5oz  
☐ 2oz  
☐ 2.5oz  
☐ 3oz

Differential Layer  
☐ Edge Cpld Ext  
☐ Edge Cpld Int Sym  
☒ Edge Cpld Int Asym  
☐ Edge Cpld Embed  
☐ Broad Cpld Shld  
☐ Broad Cpld NShld

Units  
☒ Imperial  
☐ Metric

Substrate Options  
**Custom**

Er  
**4.45**

Tg (°C)  
**130**

Temp Rise (°C)  
**20**

Temp in (°F) = 36.0

Ambient Temp (°C)  
**22**

Temp in (°F) = 71.6

Print

Solve!

Information

Total Copper Thickness  
1.40 mils

W/H = 0.181  
S/H = 0.904  
Lsat = 251.72 mils

 **SATURN**  
PCB DESIGN, INC  
Turnkey Electronic Engineering Solutions

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